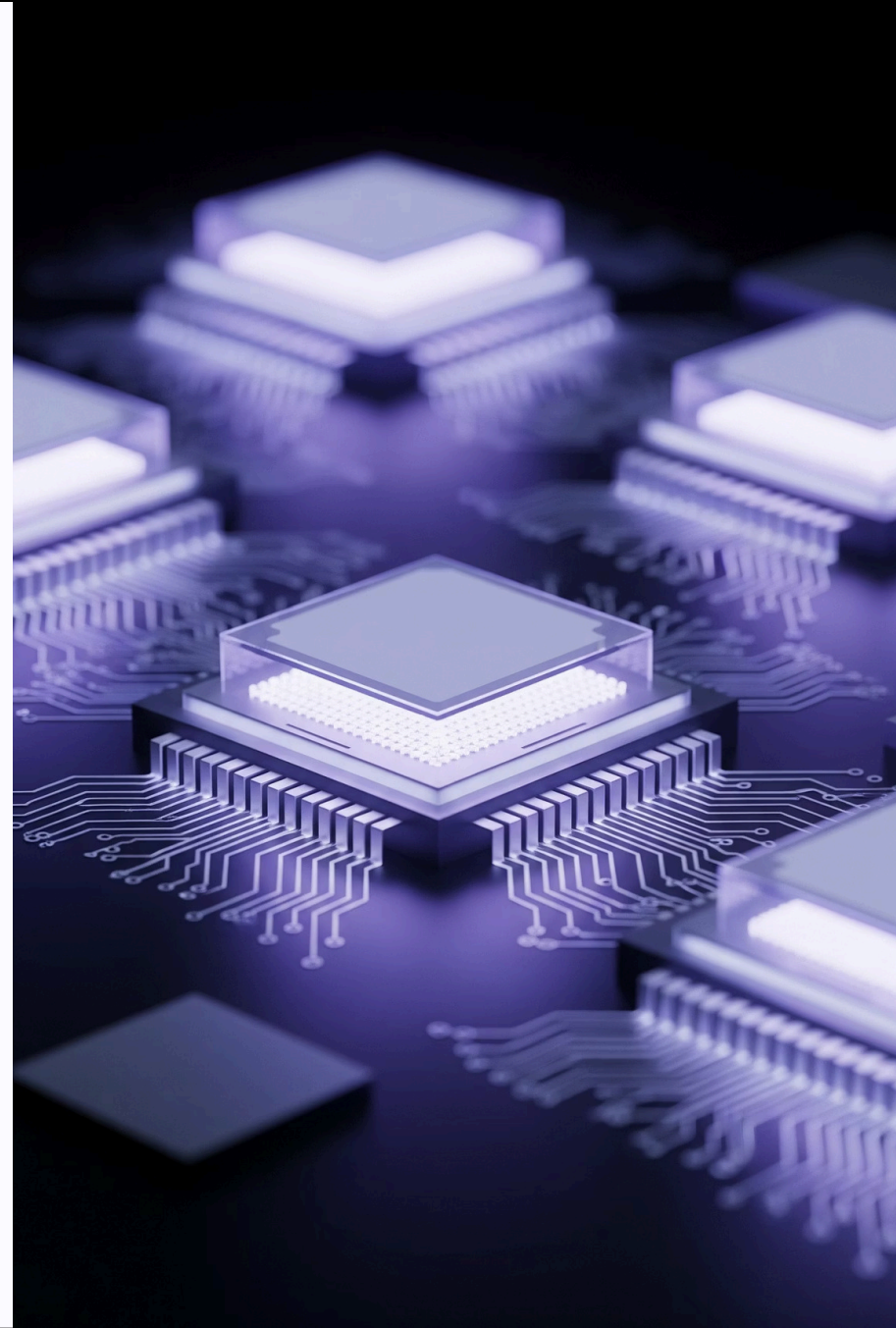


Trends in AI Chips: GPUs, TPUs, NPUs

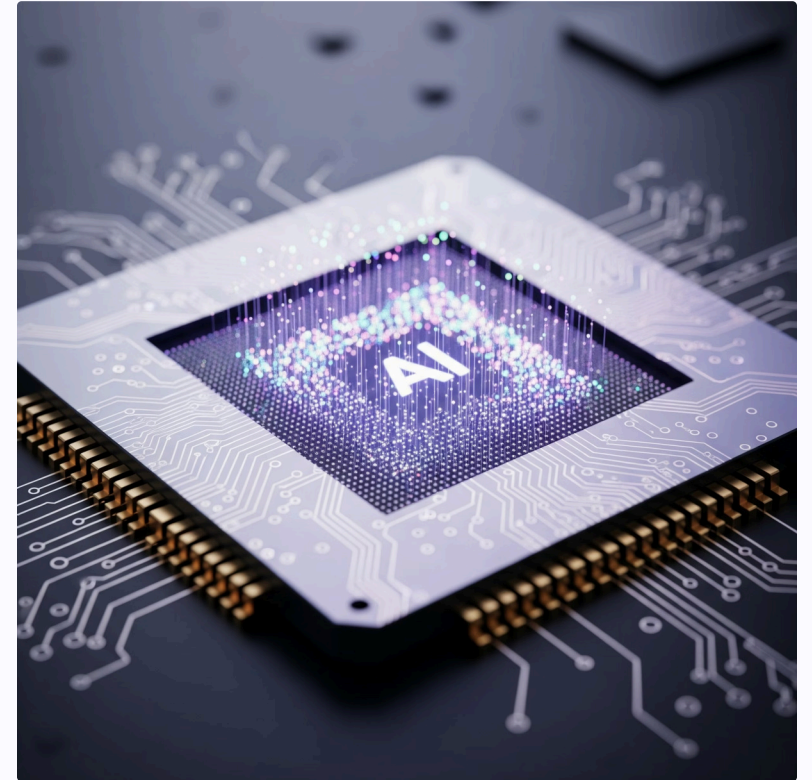
The engine of modern AI infrastructure is evolving from general-purpose GPUs toward domain-specific accelerators, with a future that will be increasingly heterogeneous, specialized, and efficient.



Why Chips Matter for AI Infrastructure

AI workloads fundamentally operate as **matrix multiplications at scale**, demanding extraordinary computational power:

- Training requires petaflops to exaflops of compute
- Inference handles billions of daily requests requiring low latency
- The evolution of chip architecture directly drives the **cost, speed, and scalability** of AI systems



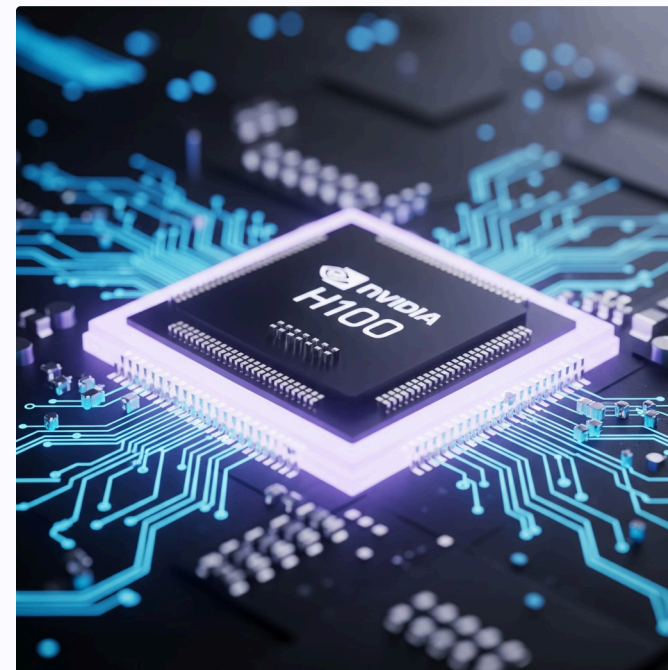
GPUs: The Workhorses of AI

Originally designed for graphics rendering, GPUs have become the **primary engines** powering modern AI development.

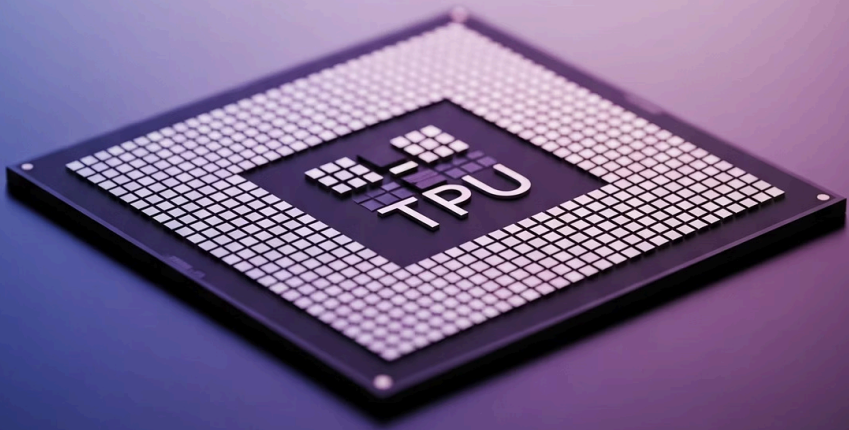
Key Strengths:

- Highly parallel architecture with thousands of CUDA cores
- Extensive ecosystem (NVIDIA CUDA, cuDNN, PyTorch support)
- Versatility across diverse AI workloads

Examples: NVIDIA A100, H100



Challenges: Expensive acquisition costs, high power consumption, and persistent supply constraints



TPUs: Google's Tensor Processing Units

Purpose-Built ASICs

Application-Specific Integrated Circuits designed exclusively for machine learning workloads

Google's AI Backbone

Powers Google products and DeepMind's breakthrough research

Matrix Operation Masters

Architecturally optimized for the tensor operations that dominate AI computations

Software Integration

Deeply integrated with JAX + TensorFlow + XLA compilation stack

TPUv5e offers a balanced performance/cost profile for both training and inference workloads, with exceptional energy efficiency at scale

NPU: Neural Processing Units

Embedded AI accelerators specifically designed for **edge and mobile computing** applications:

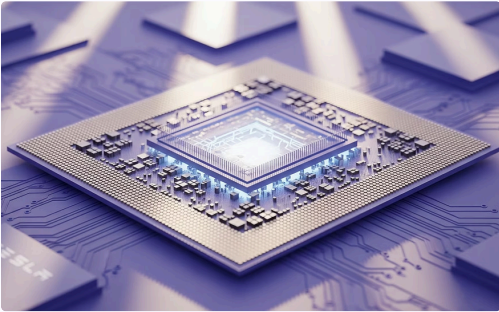
- Apple Neural Engine (A-series, M-series chips)
- Qualcomm Hexagon DSP
- Huawei Ascend

Key Benefits:

- Enables on-device inference for privacy protection
- Operates with extremely low power consumption
- Supports the growing trend of **AI at the edge** for real-time applications

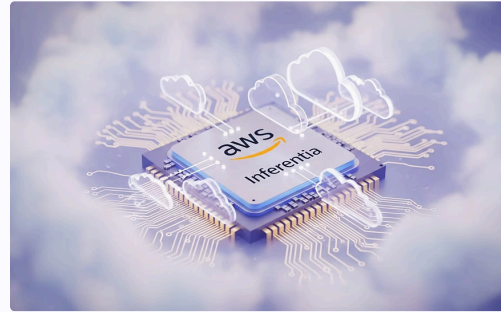


The AI Silicon Arms Race



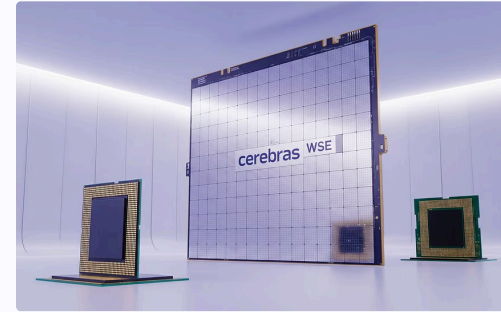
Tesla Dojo (D1)

Vision AI training at fleet scale for autonomous vehicles



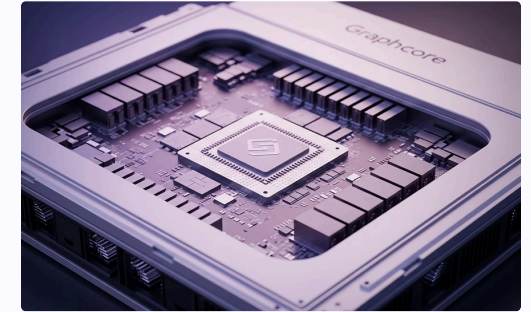
AWS Inferentia & Trainium

Cloud-native accelerators optimized for AWS infrastructure



Cerebras WSE

Wafer-scale integration enabling massive parallelism

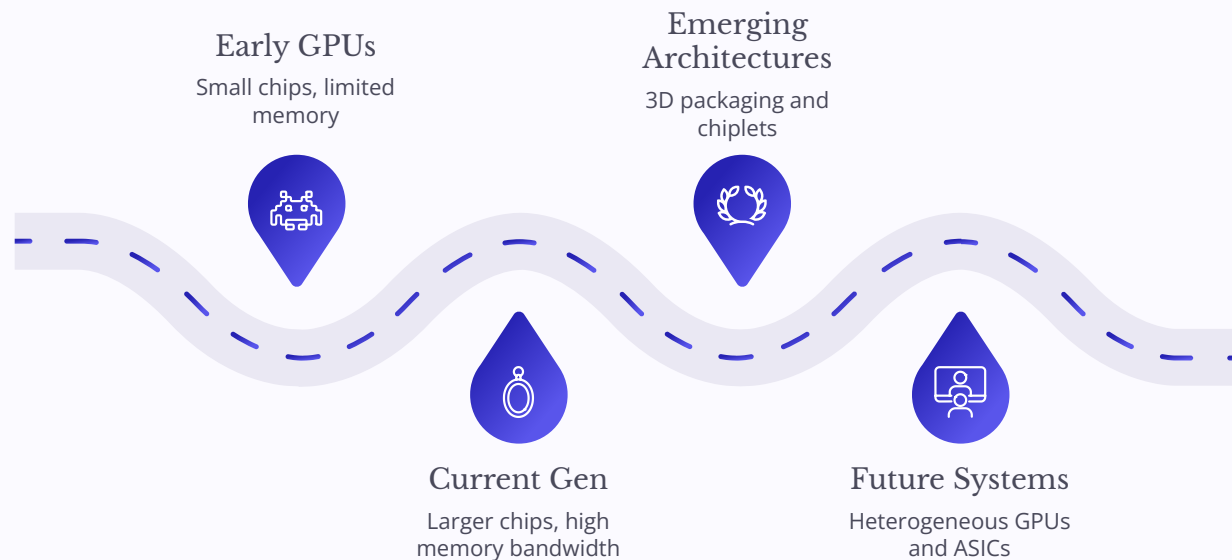


Graphcore IPU

Intelligent Processing Unit for fine-grained ML workloads

The landscape is increasingly diverse as companies develop custom silicon to gain competitive advantages in AI capabilities

Trends in Training Chips



- Increasing chip size with dramatically expanded memory bandwidth
- Adoption of 3D packaging and chiplet architectures for manufacturing scalability
- Development of specialized interconnects (NVLink, InfiniBand, optical links)
- **Heterogeneous compute**: Strategic combination of GPUs and ASICs for optimal workload handling

Trends in Inference Chips

Shifting Priorities

- Energy efficiency now prioritized over raw computational power
- **Quantization-friendly** designs supporting INT8/FP8 operations
- Edge-first NPUs targeting AR/VR, wearables, and robotics
- Cloud inference accelerators offering cost advantages over GPUs



The inference landscape is diversifying across edge and cloud environments, with specialized silicon for each deployment context

Green AI & Sustainability Trends



Precision Optimization

Shifting to lower precision formats (BF16, FP8, INT4) dramatically reduces power requirements while maintaining acceptable accuracy



Efficient Architectures

Purpose-built NPUs and ASICs deliver orders of magnitude better performance-per-watt than general-purpose processors



Advanced Cooling

Data center innovations including liquid cooling and immersion technologies enable higher density and energy efficiency

As AI compute demands grow exponentially, addressing the environmental impact has become a critical industry concern

Case Study: NVIDIA H100



The Current AI Compute Leader

- 80 billion transistors based on the Hopper architecture
- Supports FP8 precision for significant training efficiency gains
- NVLink Switch technology enables massive 256 GPU clusters
- Powers frontier large language model training at organizations like OpenAI and Anthropic

The H100 represents NVIDIA's continued dominance in high-performance AI compute, driving the largest and most capable AI models being developed today

Case Study: Google TPUv5e

Optimized for Efficiency

Google's TPUv5e represents their most cost-efficient AI accelerator designed for both training and inference workloads:

- Specifically optimized for JAX and TensorFlow frameworks
- Delivers up to 5x better performance-per-watt than GPUs for certain ML tasks
- Seamlessly integrated into Google Cloud TPU Pods for massive scale
- Positions Google Cloud as a cost-effective alternative for specific AI workloads



TPUs showcase the advantages of purpose-built AI accelerators in cloud environments



Case Study: Apple Neural Engine

On-Device Performance

Delivers 35+ trillion operations per second on iPhone devices, enabling complex AI processing without cloud connectivity

Feature Enablement

Powers on-device Siri, FaceID, computational photography, and real-time vision analysis

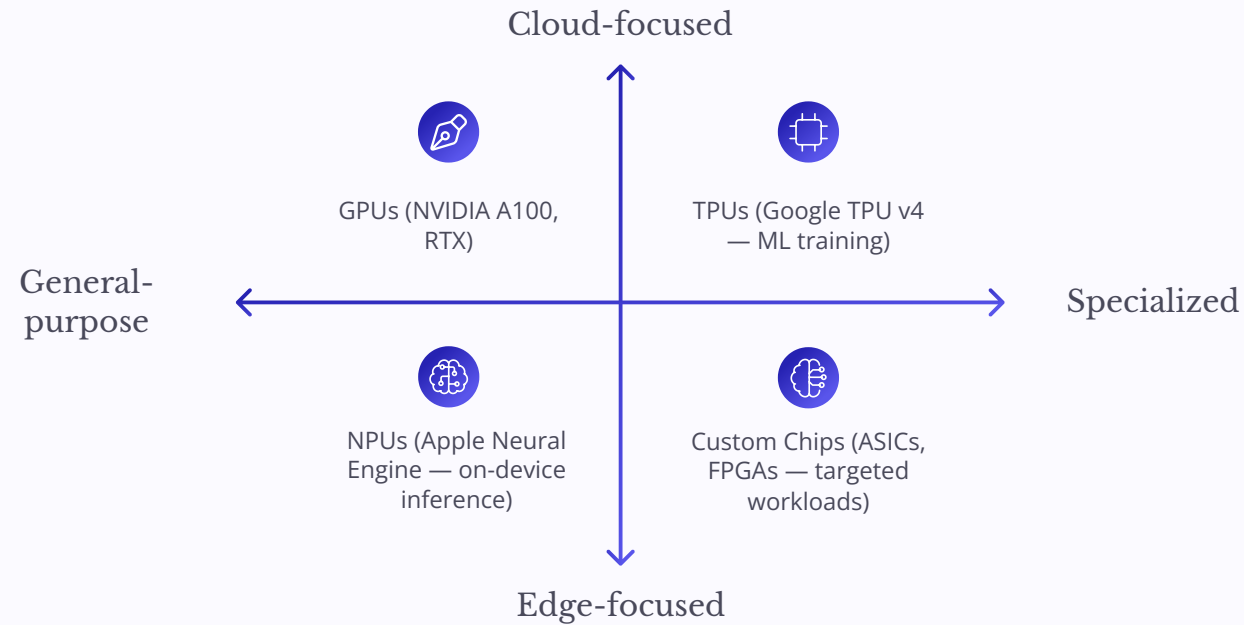
AI Democratization

Demonstrates how edge chips bring sophisticated AI capabilities to consumer devices without specialized hardware

Privacy Architecture

Ensures inference processing never leaves the device, aligning with Apple's privacy-first philosophy

The AI Compute Landscape



AI compute is shifting from general-purpose to increasingly specialized architectures

Key Lessons from Chip Trends

GPU Dominance with Rising Competition

NVIDIA's ecosystem advantage remains strong, but specialized alternatives are gaining ground in specific workloads

NPU Enable Edge AI Revolution

Edge and mobile AI accelerators are democratizing machine learning capabilities across consumer and enterprise devices

TPUs & ASICs Offer Efficiency Advantages

Purpose-built accelerators demonstrate significant performance-per-watt benefits and scaling efficiencies for focused workloads

Heterogeneous Compute Is the Future

Next-generation AI infrastructure will combine multiple accelerator types optimized for different portions of complex workloads

Discussion: Strategic AI Hardware Choices

If you were building an AI startup today, would you bet on GPUs (flexibility), TPUs (scale efficiency), or NPUs (edge focus) — and why?



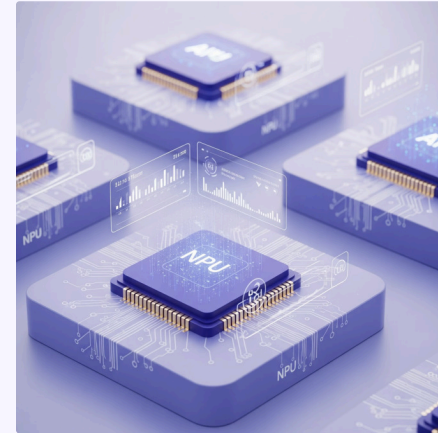
GPU Advantages

- Ecosystem maturity
- Framework compatibility
- Talent availability



TPU Advantages

- Cost efficiency at scale
- Cloud-native integration
- Energy optimization



NPU Advantages

- Privacy-preserving
- Real-time capabilities
- Disconnected operation